1.1. Escape velocity

It is tempting to use the same reasoning as in the low-orbit satellite example in Section 1.2. This reasoning gives the same result, $v = C\sqrt{gR} = C\sqrt{GM_{\rm E}/R}$, where C is some number (it turns out that $C = \sqrt{2}$). Although this solution yields the correct answer, it isn't quite rigorous, in view of the footnote in the low-orbit satellite example. Because the particle isn't always at the same radius, the force changes, so it isn't obvious that we can absorb the $M_{\rm E}$ and G dependence into one quantity, g, as we did with the orbiting satellite. Let us therefore be more rigorous with the following reasoning.

The dimensionful quantities in the problem are $[m] = M$, the radius of the earth $[R] = L$, the mass of the earth $[M_{\rm E}] = M$, and Newton's gravitational con-

stant $[G] = L^3/MT^2$. These units for G follow from the gravitational force law, $F = Gm_1m_2/r^2$. If we use no information other than these given quantities, then there is no way to arrive at the speed of $C\sqrt{GM_E/R}$, because for all we know, there could be a factor of $(m/M_E)^7$ in the answer. This number is dimensionless, so it wouldn't mess up the units.

If we want to make any progress in this problem, we have to use the fact that the gravitational force takes the form of GM_Em/r^2. This then implies (as was stated in the problem) that the acceleration is independent of m. And since the path of the particle is determined by its acceleration, we see that the answer can't depend on m. We are therefore left with the quantities G, R, and M_E, and you can show that the only combination of these quantities that gives the units of speed is $v = C\sqrt{GM_E/R}$.

1.2. Mass in a tube

The dimensionful quantities are $[g] = L/T^2$, $[\ell] = L$, $[m] = M$, and $[M] = M$. We want to produce a dimensionless number η. Since g is the only constant involving time, η cannot depend on g. This then implies that η cannot depend on ℓ, which is the only length remaining. Therefore, η depends only on m and M (and furthermore only on the ratio m/M, since we want a dimensionless number). So the answer to the stated problem is, "No."

It turns out that you have to solve the problem numerically if you actually want to find η (see Problem 8.5). Some results are: If $m \ll M$, then $\eta \approx 0.349$. If $m = M$, then $\eta \approx 0.378$. And if $m = 2M$, then $\eta \approx 0.410$.

1.3. Waves in a fluid

We want to make a speed, $[v] = L/T$, out of the quantities $[\rho] = M/L^3$, and $[B] = [F/A] = (ML/T^2)/(L^2) = M/(LT^2)$. We can play around with these quantities to find the combination that has the correct units, but let's do it the no-fail way. If $v \propto \rho^a B^b$, then we have

$$\frac{L}{T} = \left(\frac{M}{L^3}\right)^a \left(\frac{M}{LT^2}\right)^b . \tag{1.21}$$

Matching up the powers of the three kinds of units on each side of this equation gives

$$M:\ 0 = a + b, \qquad L:\ 1 = -3a - b, \qquad T:\ -1 = -2b. \tag{1.22}$$

The solution to this system of equations is $a = -1/2$ and $b = 1/2$. Therefore, our answer is $v \propto \sqrt{B/\rho}$. Fortunately, there was a solution to this system of three equations in two unknowns.

1.4. Vibrating star

We want to make a frequency, $[\nu] = 1/T$, out of the quantities $[R] = L$, $[\rho] = M/L^3$, and $[G] = L^3/(MT^2)$. These units for G follow from the gravitational force law, $F = Gm_1m_2/r^2$. As in the previous problem, we can play around with these quantities to find the combination that has the correct units, but let's do it the no-fail way. If $\nu \propto R^a\rho^bG^c$, then we have

$$\frac{1}{T} = L^a\left(\frac{M}{L^3}\right)^b\left(\frac{L^3}{MT^2}\right)^c. \tag{1.23}$$

Matching up the powers of the three kinds of units on each side of this equation gives

$$M:\ 0 = b - c, \qquad L:\ 0 = a - 3b + 3c, \qquad T:\ -1 = -2c. \tag{1.24}$$

The solution to this system of equations is $a = 0$, and $b = c = 1/2$. Therefore, our

answer is $v \propto \sqrt{\rho G}$. So it turns out that there is no R dependence.

REMARK: Note the difference in the given quantities in this problem (R, ρ, and G) and the ones in Exercise 1.12 (R, ρ, and S). In this problem with the star, the mass is large enough so that we can ignore the surface tension, S. And in Exercise 1.12 with the drop, the mass is small enough so that we can ignore the gravitational force, and hence G. ♣

1.5. **Damping**

(a) The constant b has units $[b] = [\text{Force}][v^{-n}] = (ML/T^2)(T^n/L^n)$. The other quantities are $[m] = M$ and $[V] = L/T$. There is also n, which is dimensionless. You can show that the only combination of these quantities that has units of T is

$$t = f(n)\frac{m}{bV^{n-1}}, \tag{1.25}$$

where $f(n)$ is a dimensionless function of n.

For $n = 0$, we have $t = f(0)\, mV/b$. This increases with m and V, and decreases with b, as it should.

For $n = 1$, we have $t = f(1)\, m/b$. So we *seem* to have $t \sim m/b$. This, however, cannot be correct, because t should definitely grow with V. A large initial speed V_1 requires some nonzero time to slow down to a smaller speed V_2, after which time we simply have the same scenario with initial speed V_2. So where did we go wrong? After all, dimensional analysis tells us that the answer *does* have to look like $t = f(1)\, m/b$, where $f(1)$ is a numerical factor. The resolution to this puzzle

is that $f(1)$ is infinite. If we worked out the problem using $F = ma$, we would encounter an integral that diverges. So for any V, we would find an infinite t. [6]

Similarly, for $n \geq 2$, there is at least one power of V in the denominator of t. This certainly cannot be correct, because t should not decrease with V. So $f(n)$ must likewise be infinite for all of these cases.

The moral of this exercise is that sometimes you have to be careful when using dimensional analysis. The numerical factor in front of your answer nearly always turns out to be of order 1, but in some strange cases it turns out to be 0 or ∞.

REMARK: For $n \geq 1$, the expression in Eq. (1.25) still has relevance. For example, for $n = 2$, the $m/(Vb)$ expression is relevant if you want to know how long it takes to go from V to some final speed V_f. The answer involves $m/(V_f b)$, which diverges as $V_f \to 0$. ♣

(b) You can show that the only combination of the quantities that has units of L is

$$\ell = g(n)\frac{m}{bV^{n-2}}, \tag{1.26}$$

where $g(n)$ is a dimensionless function of n.

For $n = 0$, we have $\ell = g(0)\, mV^2/b$. This increases with V, as it should.

For $n = 1$, we have $\ell = g(1)\, mV/b$. This increases with V, as it should.

For $n = 2$ we have $\ell = g(2)\, m/b$. So we *seem* to have $\ell \sim m/b$. But as in part (a), this cannot be correct, because ℓ should definitely depend on V. A large initial speed V_1 requires some nonzero distance to slow down to a smaller speed V_2, after which point we simply have the same scenario with initial speed V_2. So, from the reasoning in part (a), the total distance is infinite for $n \geq 2$, because the function g is infinite.

REMARK: Note that for integral $n \neq 1$, t and ℓ are either both finite or both infinite. For $n = 1$, however, the total time is infinite, whereas the total distance is finite. This situation actually holds for $1 \leq n < 2$, if we want to consider fractional n. ♣

1.6. **Projectile distance**

All of the possible answers have the correct units, so we'll have to figure things out by looking at special cases. Let's look at each choice in turn:

$\frac{gh^2}{v^2}$: Incorrect, because the answer shouldn't be zero for $h = 0$. Also, it shouldn't grow with g. And even worse, it shouldn't be infinite for $v \to 0$.

$\frac{v^2}{g}$: Incorrect, because the answer should depend on h.

$\sqrt{\frac{v^2 h}{g}}$: Incorrect, because the answer shouldn't be zero for $h = 0$.

$\frac{v^2}{g}\sqrt{1+\frac{2gh}{v^2}}$: Can't rule this out, and it happens to be the correct answer.

$\frac{v^2}{g}\left(1+\frac{2gh}{v^2}\right)$: Incorrect, because the answer should be zero for $v \to 0$. But this expression goes to $2h$ for $v \to 0$.

$\frac{v^2/g}{1-\frac{2gh}{v^2}}$: Incorrect, because the answer shouldn't be infinite for $v^2 = 2gh$.

[6] The total time t is actually undefined, because the particle never comes to rest. But t does grow with V, in the sense that if t is defined to be the time to slow down to some certain small speed, then t grows with V.

1.7. Two masses, one swinging

As in Section 1.4, we'll write a Maple program. We'll let `q` stand for θ, and we'll use the notation where `q1` stands for $\dot{\theta}$, and `q2` stands for $\ddot{\theta}$. Likewise for r. We'll run the program for as long as $r < 2$. As soon as r exceeds 2, the program will stop and print the value of the time.

```
r:=1:                                   # initial r value
r1:=0:                                  # initial r velocity
q:=3.14/18:                             # initial angle
q1:=0:                                  # initial angular velocity
e:=.001:                                # small time interval
i:=0:                                   # i counts the number of time steps
while r<2 do                            # run the program until r=2
```

```
i:=i+1:                                  # increase the counter by 1
r2:=(r*q1^2-9.8*(1-cos(q)))/2: # the first of the given eqs
r:=r+e*r1:                               # how r changes, by definition of r1
r1:=r1+e*r2:                             # how r1 changes, by definition of r2
q2:=-2*r1*q1/r-9.8*sin(q)/r:    # the second of the given eqs
q:=q+e*q1:                               # how q changes, by definition of q1
q1:=q1+e*q2:                             # how q1 changes, by definition of q2
end do:                                  # the Maple command to stop the do loop
i*e;                                     # print the value of the time
```

This yields a time of $t = 8.057$ s. If we instead use a time interval of 0.0001 s, we obtain $t = 8.1377$ s. And a time interval of 0.00001 s gives $t = 8.14591$ s. So the correct time must be somewhere around 8.15 s.